

# Homework 5

Due September 30, 2025.

## Problem 1.

Show that in odd dimensions (Euclidean or Minkowskian) both irreps of the Dirac matrices are real or pseudoreal if  $C = C_+$ , but if  $C = C_-$  then the irreps are complex. (*Hint:* For  $C_+$  this is obvious. For  $C_-$  show that if  $\gamma^{\mu,T} = -C_- \gamma^\mu C_-^{-1}$ , then there is no matrix  $S$  such that also  $\gamma^{\mu,T} = S \gamma^\mu S^{-1}$ .)

## Problem 2.

Show that one can always rescale  $C_+$  and  $C_-$  such that they become unitary matrices. (See the notes for a hint.)

## Problem 3.

Complete in table 11.2 on page 153 the entries for the case  $6 + 1$ .

**Comment:** These properties will be used when we discuss the spinor irreps of orthogonal Lie algebras.

## Problem 4.

Show that if in dimension  $d = s + t$  no Majorana spinor exists, a symplectic Majorana spinor (defined in the notes) satisfies the consistency condition for a symplectic Majorana spinor to exist.

## Bonus Problem.

When we constructed particular irreps for the Dirac matrices in arbitrary dimensions, we justified this by claiming that many of the properties of Dirac matrices are independent of a particular representation. We now prove this claim. Show that

- (i) if  $C_+$  or  $C_-$  is symmetric or antisymmetric, then it remains so after a similarity transformation.
- (ii) the reality properties of an irrep (real, pseudoreal, or complex) do not change by a similarity transformation.